

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 6_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Elsa subscribes to a premium service with a base monthly cost, a service tax and an extra feature cost. Assist her in writing an inheritance program that takes input for these values and calculates the total monthly cost.

Refer to the below class diagram:

Input Format

The first line of input consists of a double value, representing the base monthly cost.

The second line consists of a double value, representing the service tax.

The third line consists of a double value, representing the extra feature cost.

Output Format

The output prints "Rs. X" where X is a double value, rounded off to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10.0

2.5

5.0

Output: Rs. 17.50

Answer

```
import java.util.Scanner;
```

```
// You are using Java
```

```
// Parent class
```

```
class BasePlan {
```

```
    protected double baseCost;
```

```
    protected double serviceTax;
```

```
    BasePlan(double baseCost, double serviceTax) {
```

```
        this.baseCost = baseCost;
```

```
        this.serviceTax = serviceTax;
```

```
    }
```

```
    public double calculateBaseTotal() {
```

```
        return baseCost + serviceTax;
```

```
    }
```

```
}
```

```
// Child class
```

```
class PremiumSubscription extends BasePlan {
```

```
    private double extraFeatureCost;
```

```
    PremiumSubscription(double baseCost, double serviceTax, double
```

```
extraFeatureCost) {  
    super(baseCost, serviceTax);  
    this.extraFeatureCost = extraFeatureCost;  
}  
  
public double calculateMonthlyCost() {  
    return calculateBaseTotal() + extraFeatureCost;  
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        double baseMonthlyCost = scanner.nextDouble();  
        double serviceTax = scanner.nextDouble();  
        double extraFeatureCost = scanner.nextDouble();  
  
        PremiumSubscription premiumSubscription = new  
PremiumSubscription(baseMonthlyCost, serviceTax, extraFeatureCost);  
  
        double totalMonthlyCost = premiumSubscription.calculateMonthlyCost();  
  
        System.out.printf("Rs. %.2f%n", totalMonthlyCost);  
  
        scanner.close();  
    }  
}
```

Status : Correct

Marks : 10/10